CHAPTER 6

Nuclear Structure

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Throughout the ages and in every civilization, people have developed explanations of observed behaviors. These explanations are based on the principle of causality, i.e. every effect has a cause and the same cause produces always the same effect. We call these explanations *models*.

Scientists are professional model-builders. Observed phenomena are used to develop a model, which then is tested through new experiments. Actually, all things we claim that

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we know today are in reality models of some unknown reality. This is continuously noted when all of a sudden a new discovery seem to falsify an older model e.g. the current discussion of the possibility that neutrinos actually travel faster than the speed of light. If these experiments are true we have to remodel much of that which we today consider being the truth. Indirect model building is familiar to every chemist: although we cannot see the atoms and molecules which we add into a reaction vessel, we certainly have some idea about what is going to happen. It is indeed our enjoyment in developing models which causes us to experiment in science. We also want to be able to make quantitative predictions based on our models which we therefore formulate in mathematical terms. To allow tractable calculations most models involves simplifications of the "real world". Of course, since man is fallible some models may turn out to be wrong but as new data accumulate, wrong or naive models are replaced by better ones.

We have already shown how one model for the nuclear structure, the liquid drop model, has helped us to explain a number of nuclear properties, the most important being the shape of the stability valley. But the liquid drop model fails to explain other important properties. In this chapter we shall try to arrive at a nuclear model which takes into account the quantum mechanical properties of the nucleus based mainly on the interactions and properties of neutrons and protons. Naturally it is also possible to go even deeper and discuss the nuclear structure through the interactions of quarks or using the string theory but that is beyond the scope of this book and we instead refer to textbooks in nuclear physics.

6.1. REQUIREMENTS OF A NUCLEAR MODEL

Investigation of light emitted by excited atoms (J. Rydberg 1895) led N. Bohr to suggest the quantized model for the atom, which became the foundation for explaining the chemical properties of the elements and justifying their ordering in the periodic system. From studies of molecular spectra and from theoretical quantum and wave mechanical calculations, we are able to interpret many of the most intricate details of chemical bonding.

In a similar manner, patterns of nuclear stability, results of nuclear reactions and spectroscopy of radiation emitted by nuclei have yielded information which helps us develop a picture of nuclear structure. But the situation is more complicated for the nucleus than for the atom. In the nucleus there are two kinds of particles, protons and neutrons, packed close together, and there are two kinds of forces - the electrostatic force and the short range strong nuclear force. This more complex situation has caused slow progress in developing a satisfactory model, and no single nuclear model has been able to explain all the nuclear phenomena.

6.1.1. Some general nuclear properties

Let us begin with a summary of what we know about the nucleus, and see where that leads us.